



March 2026

**COLLECTIVE LAB X GDG BUILDATHON**

# **BUILDATHON SOLUTION PROPOSAL**

**TARGET UNIT: COUNCELLING SERVICES**

**CLINICAL MANAGEMENT PORTAL**

**TEAM NAME: TEAM E**

# **1. PROJECT & CONTEXT**

## **PROBLEM STATEMENT**

The department relies on manual, walk-in based appointment booking and physical session documentation, leading to administrative delays and heavy reporting workloads

## **CORE OPERATIONAL BOTTLENECK**

Manual scheduling conflicts and the time-intensive process of converting handwritten session drafts into structured clinical reports.

## **WHO IS AFFECTED?**

Students seeking mental health support and the department's therapists.

## **MAIN PAIN POINTS**

Requirement for physical presence to book, manual time assignment, and tedious report writing post-session.

## 2. SOLUTION CONCEPT

### PROPOSED SOLUTION

A Clinical Management Portal featuring automated self-service scheduling and an AI-powered documentation assistant.

### KEY FEATURES:

**Self-Service Booking:** Real-time availability calendar for students.

**AI SOAP(Subjective, Objective, Assessment, and Plan) Note Generator:** Converts therapist drafts into structured clinical reports using LLM processing.

**Crisis Flagging System:** Scans intake forms for urgent keywords to prioritize high-risk cases.

**How it Solves the Problem?** It removes the need for physical office visits for booking and cuts documentation time by up to 70% through AI-assisted formatting.

# 3. TECHNICAL IMPLEMENTATION PLAN

| LAYER           | Tool        | Purpose                                                                                    |
|-----------------|-------------|--------------------------------------------------------------------------------------------|
| Frontend        | Lovable     | AI full-stack builder for rapid UI prototyping<br>React/Next.js                            |
| Backend Logic   | Antigravity | AI full-stack builder for rapid backend logic<br>Next.js                                   |
| Database & Auth | Github      | PostgreSQL, authentication, real-time subscriptions,<br>Edge Functions, Row Level Security |
| Deployment      | Vercel      | Continuous deployment from GitHub – live URL                                               |

## 4. SUCCESS CRITERIA & MILESTONES

### MVP SUCCESS DEFINITION

- A "Single Source of Truth" dashboard where the Desk Officer can see the availability of all 5 session categories at once.
- Zero-conflict booking logic: The system must programmatically block a counselor's time across all roles once a single slot is taken.
- Successful implementation of a "Walk-in to Digital" queue that notifies the relevant counselor in real-time.

### KEY MILESTONES CHECK-INS

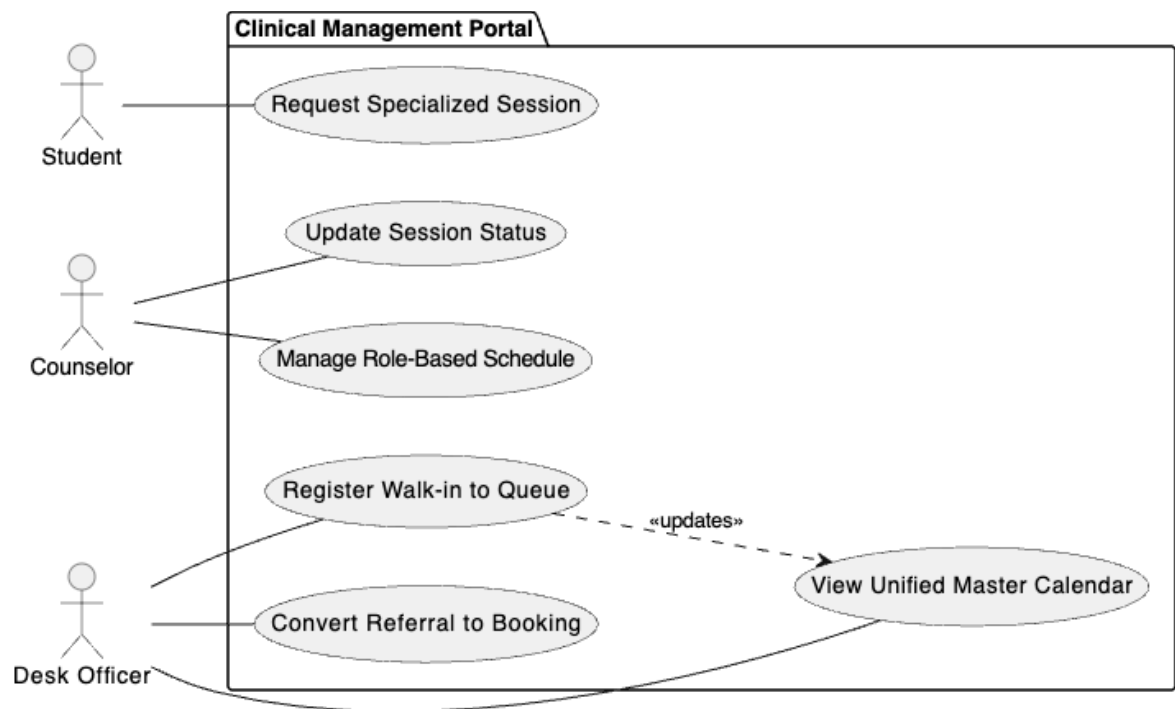
- **Week 1-2:** Development of the "Conflict Resolver" algorithm to handle overlapping session types.
- **Week 3-4:** Frontend dashboard completion using React to visualize the multi-role calendar.
- **Week 5:** Stress-testing the system with concurrent booking requests to ensure no double-bookings occur.

## 5. USER INTERACTION

The Clinical Management Portal serves as a secure bridge between students and mental health professionals. The use case diagram illustrates a shift from physical dependency to a digital-first workflow.

- **Student Interaction:** Students independently access the system to view real-time availability and secure appointments without visiting the office.
- **Therapeutic Efficiency:** Therapists interact with an AI-driven backend that processes raw session notes into structured clinical reports (SOAP notes), drastically reducing administrative overhead.
- **Automated Oversight:** The system acts as an autonomous monitor, scanning intake data to flag high-risk cases for the Administrator, ensuring immediate crisis response.

## 6. UML DIAGRAM



# Thank You

submitted by Daniel Johnson